

Distress and safety

Combined factor-model portfolio test

Practitioner factor-model runner

2026-08-26T14:04:11+00:00

Abstract

Model. Distress and safety, one long-short book from 3 sleeves (Solvency, Liquidity, EarningsStability). 5857 symbols (full screened universe), 2008-12-01 to 2026-07-01 (212 months).

Earned. -0.17% annual arithmetic return (-0.43% compounded), 7.16% volatility, Sharpe -0.02, HAC t=-0.09.

Survives? vs SPY market benchmark: alpha 0.46%, t=0.19. vs VAL+MOM+BAB+QMJ: alpha -0.56%, t=-0.41. Conventional bar is $|t| \geq 2$.

Best swap. (Standard model, Liquidity for QMJ) improves: Sharpe 0.07 against -0.02 (+0.09).
Verdict. Reject for now: it clears neither control at $|t| \geq 2$.

Exploratory model diagnostic from a single configured run, not evidence from an untouched confirmatory sample.

1 Research objective

Test whether the configured model earns a useful next-month return and whether the benchmark or standard factors explain it. Signals are observed at month-end; returns are next-calendar-month simple returns.

Construction. Normalize within month, combine components within sleeves, then combine sleeves at the stock level using Solvency 33%, Liquidity 33%, Earnings Stability 33%. Construct one portfolio; apply beta neutrality and costs once. Sleeve returns are not averaged.

2 Configuration and sample

Factor	Composite signals	Signal transform	Composite transform	Winsor	Min inputs	Mean inputs
Solvency	altman_z, debt_to_assets	rank	rank	1%-99%	2	2.0
Liquidity	current_ratio, cash_ratio	rank	rank	1%-99%	2	2.0
Earnings Stability	roe_volatility	rank	rank	1%-99%	1	1.0

Model construction	Book	Beta neutral	Bucket frac.	Cost bps	Min factors	Benchmark
signal weighted	long-short	no	0.33	0.0	3	SPY

Portfolio	Mean stocks	Minimum stocks
Combined model	815.8	437
Solvency	776.6	430
Liquidity	835.2	504
Earnings Stability	756.1	50

Sample: 5,857 symbols, 195,258 stock-months, 2001-10-01–2026-08-01. Screens: price 1.0, 21-day dollar volume None, market cap None, top-cap fraction 0.375, exclude financials True. Benchmark: SPY. Factor controls: standard_recreated_factors.csv.

3 Portfolio returns

Raw net performance on one shared window. Return and volatility use monthly arithmetic moments; t is HAC.

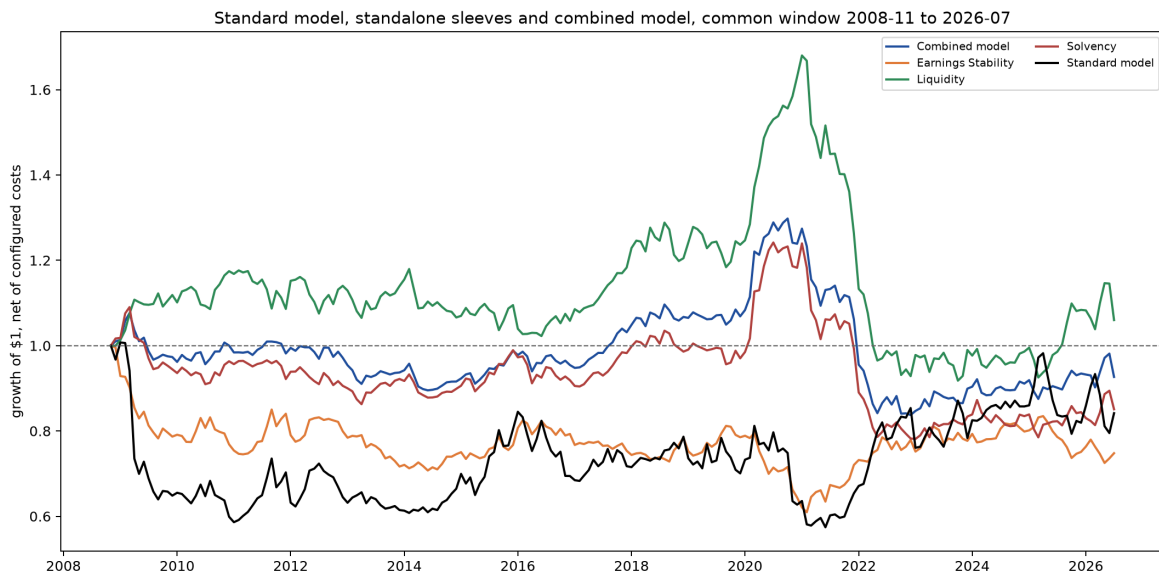


Figure 1: Growth of one dollar in each book, net of configured costs, on the common window. The path begins at one before the first return month.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	212	-0.17%	-0.43%	7.16%	-0.02	-0.09
Solvency	212	-0.59%	-0.91%	8.05%	-0.07	-0.27
Liquidity	212	0.74%	0.33%	9.09%	0.08	0.30
Earnings Stability	212	-1.34%	-1.63%	7.79%	-0.17	-0.70
Standard model	212	0.08%	-0.97%	14.30%	0.01	0.02

4 Benchmark fit and benchmark-orthogonalized returns

Regress each raw net return on SPY market benchmark. R^2 is variance explained; alpha is the annualized intercept; t and F use HAC inference. This is separate from standard-factor attribution. Estimated MKT loading: -0.043. Annual raw return -0.17% = controlled alpha 0.46% + benchmark contribution -0.64%. Orthogonal means zero sample covariance with the benchmark, not a lower mean. This is an ex-post regression diagnostic, not beta-neutral construction or a tradeable improvement.

Portfolio	N	Loading (MKT)	t (loading)	R^2 (SPY market benchmark)	F
Combined model	212	-0.04	-0.77	0.008	0.59
Earnings Stability	212	-0.27	-6.34	0.270	40.17
Liquidity	212	0.20	3.48	0.104	12.12
Solvency	212	-0.02	-0.28	0.001	0.08
Standard model	228	-0.56	-9.19	0.376	84.50

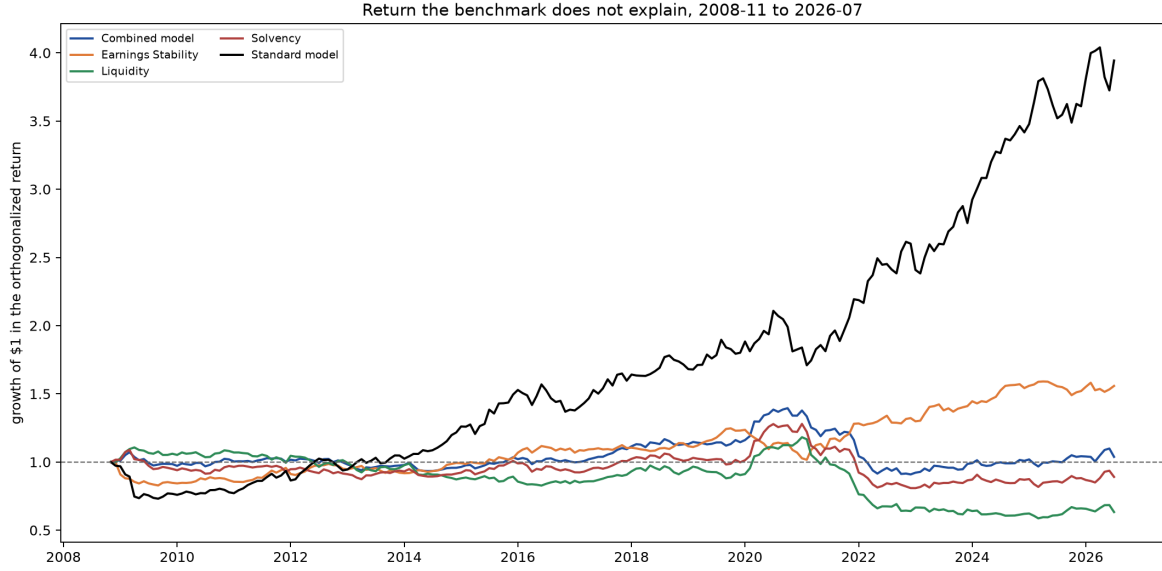


Figure 2: Returns orthogonalized to the SPY market benchmark: growth of one dollar in each book once the SPY market benchmark component is removed. A flat path earns nothing the benchmark does not already deliver.

The table summarizes the residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	212	0.46%	0.21%	7.13%	0.07	0.24
Earnings Stability	212	2.73%	2.54%	6.66%	0.41	1.62
Liquidity	212	-2.21%	-2.55%	8.60%	-0.26	-0.93
Solvency	212	-0.33%	-0.65%	8.05%	-0.04	-0.15
Standard model	228	8.02%	7.65%	11.18%	0.72	3.04

5 Attribution to configured factor controls

Regress each raw net return simultaneously on VAL+MOM+BAB+QMJ: $r_{p,t} = \alpha + \sum_k \beta_k f_{k,t} + \varepsilon_t$. The first table gives the loadings, with HAC t in brackets.

Portfolio	VAL	MOM	BAB	QMJ
Combined model	-0.31 (-5.21)	0.10 (2.67)	-0.09 (-1.97)	0.04 (0.85)
Earnings Stability	0.11 (2.09)	0.05 (1.14)	0.02 (0.44)	0.28 (4.81)
Liquidity	-0.34 (-4.98)	0.02 (0.35)	-0.12 (-2.23)	-0.22 (-3.81)
Solvency	-0.38 (-6.03)	0.13 (3.48)	-0.08 (-1.25)	0.04 (0.79)

Return contribution is $\beta_k \times 12 \times \overline{f_k}$. Explained plus alpha equals total annual return.

Portfolio	VAL	MOM	BAB	QMJ	Explained	Total
Combined model	0.35%	0.57%	-0.44%	0.23%	0.70%	0.14%
Earnings Stability	-0.12%	0.27%	0.11%	1.52%	1.77%	-1.44%
Liquidity	0.38%	0.11%	-0.58%	-1.20%	-1.29%	1.17%
Solvency	0.43%	0.72%	-0.37%	0.22%	0.99%	-0.32%

5.1 Simultaneous factor spanning and factor-orthogonalized returns

The same regression as a spanning test. R2 is the share of monthly net-return variance jointly explained by the factor controls.

Portfolio	N	R2 (4F)	F
Combined model	211	0.427	19.81
Earnings Stability	211	0.283	8.88
Liquidity	211	0.382	14.99
Solvency	211	0.486	24.53

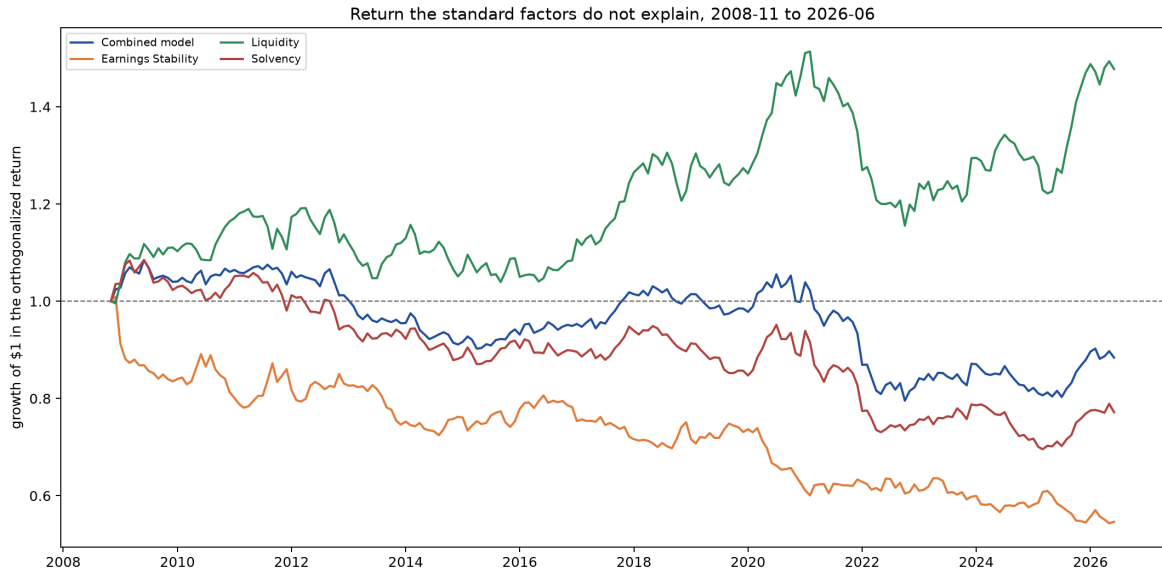


Figure 3: Returns orthogonalized to VAL+MOM+BAB+QMJ: growth of one dollar in each book once its factor exposures are removed. This is the Alpha column above, drawn as a path.

The table summarizes the factor-residual paths above.

Portfolio	Months	Ann. ret.	CAGR	Ann. vol.	Sharpe	t
Combined model	211	-0.56%	-0.70%	5.34%	-0.10	-0.43
Earnings Stability	211	-3.21%	-3.38%	6.60%	-0.49	-2.08
Liquidity	211	2.47%	2.25%	7.02%	0.35	1.38
Solvency	211	-1.31%	-1.46%	5.73%	-0.23	-0.92

5.2 The new model against the reference model

Regress the combined model on Standard model as one return series. High R2 with zero alpha means the reference reproduces the model.

Portfolio	N	Alpha	t	R2 (Standard model)	F
Combined model	212	-0.18%	-0.09	0.003	0.19

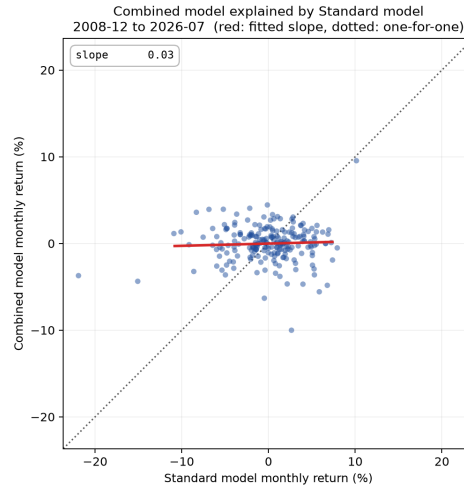


Figure 4: Each month of the combined model against Standard model, with the fitted slope. A tight cloud along the line is a model the reference reproduces; vertical spread is what it cannot.

5.3 Correlation and principal components

Check whether a sleeve duplicates a standard factor. The combined model is excluded because its overlap with its own sleeves is mechanical. PCA uses standardized sleeve and standard-factor returns over 211 common months. The share shown in the plot is the largest covariance-matrix eigenvalue divided by the sum of all eigenvalues: the fraction captured by one dominant co-movement, not return or regression R2. The combined model is excluded because its overlap with its own sleeves is mechanical.

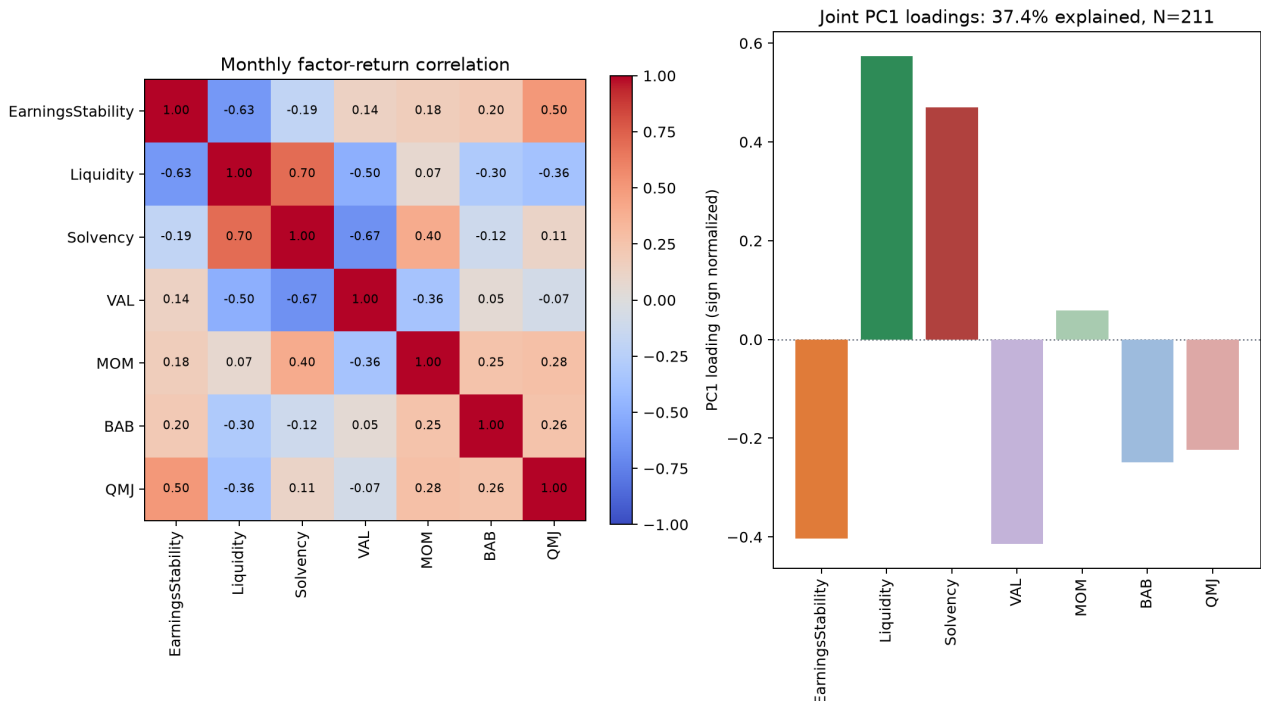


Figure 5: Monthly return correlations and sign-normalized first-principal-component loadings across the standalone sleeves and the supplied standard factors.

6 Signal persistence

Mean rank IC between formation exposure and a single later month's return. Flat profiles persist; fast decay implies faster rebalancing.

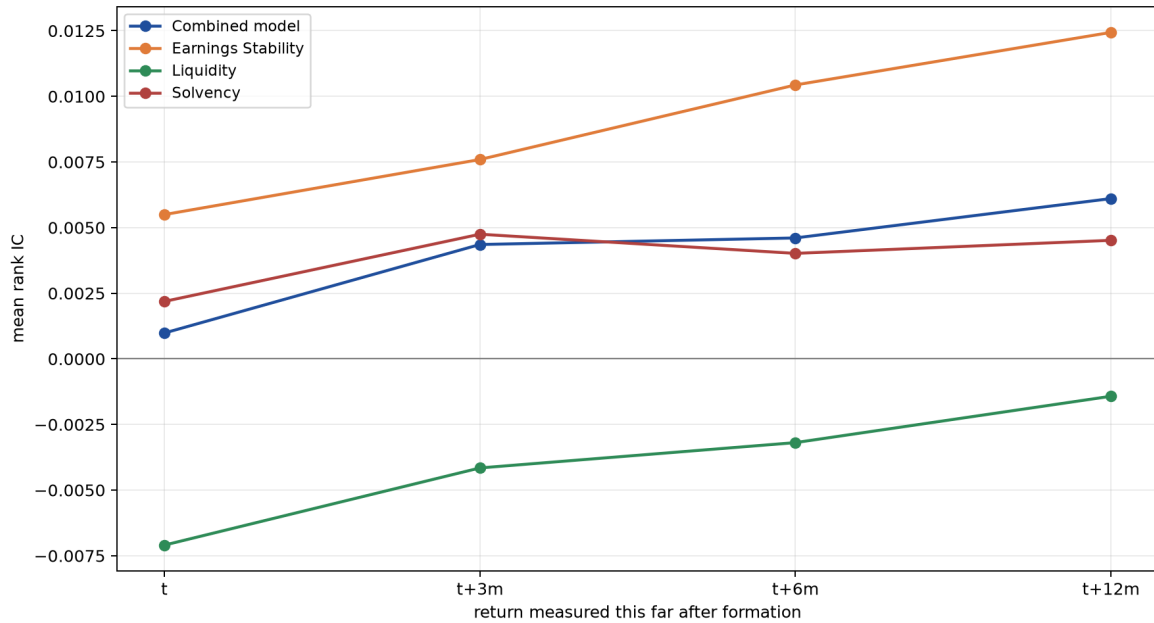


Figure 6: Mean rank information coefficient at each horizon. A line that stays flat is a signal that keeps working long after the month it is traded.

Portfolio	Horizon	Mean IC	t	Months
Combined model	t	0.0010	0.15	228
Combined model	t+3m	0.0044	0.65	225
Combined model	t+6m	0.0046	0.69	222
Combined model	t+12m	0.0061	0.93	216
Earnings Stability	t	0.0055	0.77	212
Earnings Stability	t+3m	0.0076	1.11	209
Earnings Stability	t+6m	0.0104	1.53	206
Earnings Stability	t+12m	0.0124	1.74	200
Liquidity	t	-0.0071	-0.84	228
Liquidity	t+3m	-0.0042	-0.48	225
Liquidity	t+6m	-0.0032	-0.37	222
Liquidity	t+12m	-0.0014	-0.17	216
Solvency	t	0.0022	0.30	228
Solvency	t+3m	0.0047	0.62	225
Solvency	t+6m	0.0040	0.53	222
Solvency	t+12m	0.0045	0.62	216

7 Candidate comparisons and replacement

Standalone compares candidate with incumbent. In-model replaces the incumbent sleeve and rebuilds the standard model; diversification can reverse the standalone ranking.

Each pair judged twice: as a standalone factor, and swapped into the model
dashed line: the unmodified standard model (-0.26%)

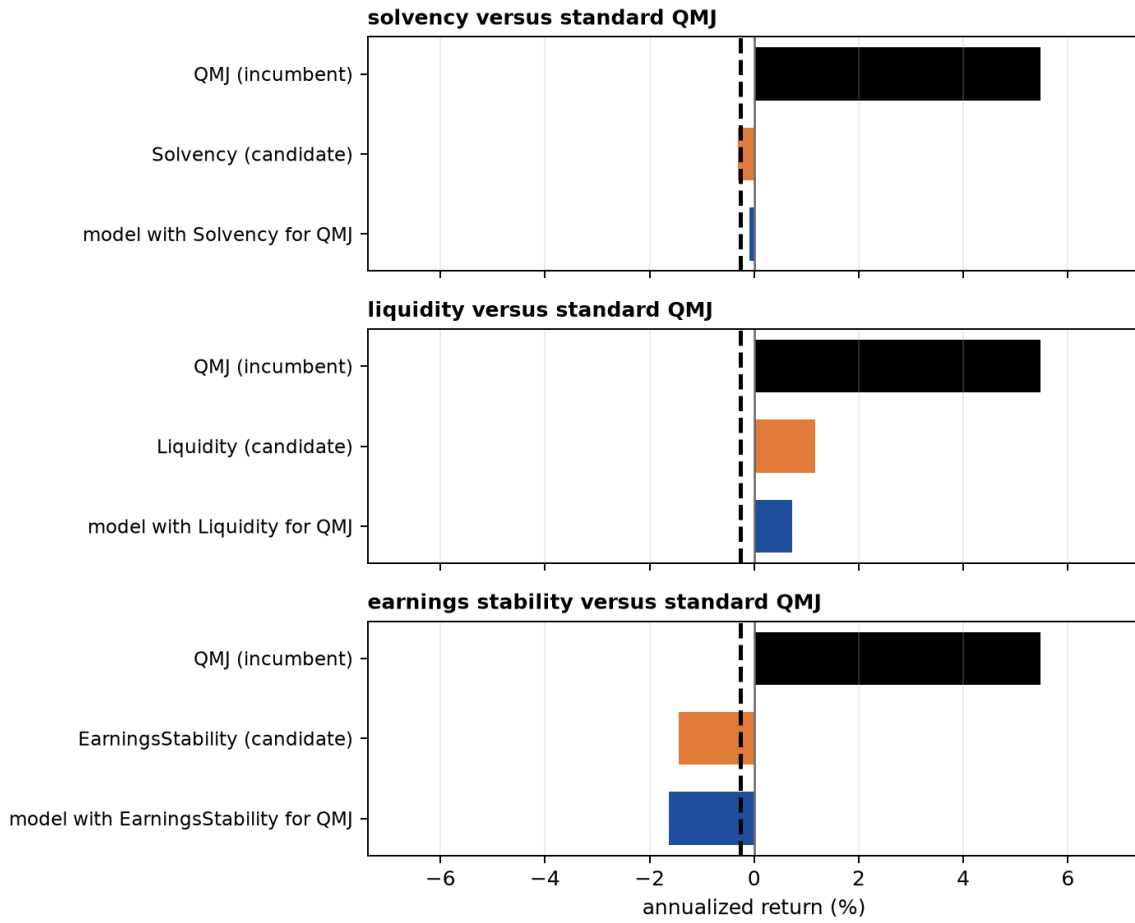


Figure 7: Each pair judged twice: the incumbent factor against the candidate standing alone, then the model with the candidate swapped in. The dashed line is the unmodified standard model.

Pair	Role	Portfolio	Ann. ret.	Change	Alpha	Change	t
solvency versus standard QMJ	incumbent	QMJ	5.48%	—	10.07%	—	3.58
	candidate in model	Solvency	-0.32%	-5.79%	-0.03%	-10.10%	-0.01
		Standard model, Solvency for QMJ	-0.09%	+0.17%	7.20%	-1.45%	3.52
liquidity versus standard QMJ	incumbent	QMJ	5.48%	—	10.07%	—	3.58
	candidate in model	Liquidity	1.17%	-4.30%	-1.76%	-11.83%	-0.62
		Standard model, Liquidity for QMJ	0.73%	+0.99%	6.72%	-1.92%	3.66
earnings stability versus standard QMJ	incumbent	QMJ	5.48%	—	10.07%	—	3.58
	candidate in model	Earnings Stability	-1.44%	-6.92%	2.65%	-7.42%	1.35
		Standard model, Earnings Stability for QMJ	-1.63%	-1.37%	6.73%	-1.91%	2.67

8 Conclusion

Verdict. Reject for now: it clears neither control at $|t| \geq 2$. vs SPY market benchmark: alpha 0.46%, $t=0.19$; vs VAL+MOM+BAB+QMJ: alpha -0.56%, $t=-0.41$.

9 Reproducibility

Run ID:

20260826T140411Z_distress-and-safety_7c0673ec61

Configuration hash: 7c0673ec61. Gold version: version=20260826T131322Z. The run directory contains configuration, data products, figures, source, log, and PDF. Re-runs never overwrite it.